Practice 1: Additive patterns

Name: ____________________ Date: __________

Find the next two terms.

1. 7, 12, 17, 22, …

Work space

2. 45, 40, 35, 30, …

Work space

3. 106, 116, 126, 136, …

Work space

Practice 2: Find the additive rule

Name: ____________________ Date: __________

State the rule and complete the sequence.

1. 3, 10, 17, 24, **,**

Work space

2. 92, 84, 76, 68, **,**

Work space

3. A row has 5 chairs. Each new row has 4 more chairs. Write the first five row totals.

Work space

Practice 3: Multiplicative patterns

Name: ____________________ Date: __________

Find the next two terms and name the rule.

1. 2, 6, 18, 54, …

Work space

2. 320, 160, 80, 40, …

Work space

3. 1, 4, 16, 64, …

Work space

Practice 4: Input-output tables

Name: ____________________ Date: __________

Find the rule, then give the missing output.

1. Rule: multiply by 3 and add 2. Find the output for input 7.

Work space

2. Inputs 1, 2, 3 give outputs 5, 8, 11. Find the output for input 6.

Work space

3. Inputs 4, 6, 8 give outputs 12, 18, 24. State the rule and find the output for 10.

Work space

Practice 5: Tables and coordinate pairs

Name: ____________________ Date: __________

Complete each table or pair.

1. Rule $y = x + 9$: find y for $x = 0, 4, 11.$

Work space

2. Rule $y = 2x$: complete $(1, \_), (3, \_), (7, \_).$

Work space

3. A table has x: 2, 5, 8 and y: 7, 16, 25. Find y when $x = 11.$

Work space

Practice 6: Write a rule from a table

Name: ____________________ Date: __________

Describe the relationship with words and an expression.

1. x: 1, 2, 3; y: 6, 8, 10.

Work space

2. x: 2, 4, 6; y: 5, 10, 15.

Work space

3. x: 0, 1, 2; y: 12, 17, 22.

Work space

Practice 7: Evaluate expressions

Name: ____________________ Date: __________

Substitute the given value for the variable.

1. Find $n + 14$ when $n = 9$.

Work space

2. Find $4m$ when $m = 7$.

Work space

3. Find $3p + 2$ when $p = 6$.

Work space

Practice 8: Write expressions

Name: ____________________ Date: __________

Write an expression for each phrase.

1. 8 more than q.

Work space

2. 5 times r.

Work space

3. 6 less than t.

Work space

Practice 9: Translate situations

Name: ____________________ Date: __________

Define a variable and write an expression.

1. A box holds b pencils. How many pencils are in 7 boxes?

Work space

2. Maya has d dollars and earns 12 more. Write her new amount.

Work space

3. A 40-page book has p pages read. Write the pages left.

Work space

Practice 10: One-step addition equations

Name: ____________________ Date: __________

Solve for the variable and check.

1. $x + 18 = 43$

Work space

2. $a + 27 = 60$

Work space

3. $15 + y = 52$

Work space

Practice 11: One-step subtraction equations

Name: ____________________ Date: __________

Solve for the variable and check.

1. $z - 16 = 29$

Work space

2. $75 - c = 38$

Work space

3. $h - 24 = 19$

Work space

Practice 12: Multiplication equations

Name: ____________________ Date: __________

Solve each equation.

1. $5x = 45$

Work space

2. $7n = 63$

Work space

3. $4p = 28$

Work space

Practice 13: Division equations

Name: ____________________ Date: __________

Solve each equation.

1. $q \div 6 = 8$

Work space

2. $72 \div r = 9$

Work space

3. $s \div 5 = 11$

Work space

Practice 14: Word problems with equations

Name: ____________________ Date: __________

Write an equation, solve, and label the answer.

1. A number plus 35 is 82. What is the number?

Work space

2. Six equal bags hold 54 apples. How many per bag?

Work space

3. Leo had some stickers, gave away 18, and has 27 left. How many did he start with?

Work space

Practice 15: Extend a pattern

Name: ____________________ Date: __________

Use the rule to find the requested term.

1. Start at 6 and add 9. Find term 6.

Work space

2. Start at 3 and multiply by 2. Find terms 5 and 6.

Work space

3. Term 1 is 12; subtract 3 each time. Find term 8.

Work space

Practice 16: Pattern reasoning

Name: ____________________ Date: __________

Explain the rule or find the missing term.

1. 11, 18, 25, 32, blank. What is term 5?

Work space

2. A sequence begins 4, 12, 36. Is it adding or multiplying? Give the next term.

Work space

3. 50, 45, 40, blank, 30. Find the missing term and rule.

Work space

Practice 17: Compare rules

Name: ____________________ Date: __________

Compare the two rules for the given input.

1. Rule A: $x + 10$; Rule B: $2x$. Which gives more when $x = 8$?

Work space

2. Rule A: $3x$; Rule B: $x + 12$. Compare when $x = 5$.

Work space

3. Rule A: $4x + 1$; Rule B: $4x + 5$. Which is always greater? Explain.

Work space

Practice 18: Real-world patterns

Name: ____________________ Date: __________

Model each situation with a rule or equation.

1. A theater row has 8 seats; each next row has 2 more. How many seats are in row 7?

Work space

2. A plant is 14 cm tall and grows 3 cm each week. How tall after 6 weeks?

Work space

3. Four notebooks cost $20. Write an equation for the price c of one notebook and solve it.

Work space

Practice 19: Mixed review

Name: ____________________ Date: __________

Show your reasoning.

1. Continue 5, 15, 45. Write the next two terms and state the rule.

Work space

2. Evaluate $2v + 7$ when $v = 9$.

Work space

3. Solve $w + 26 = 71$.

Work space

Practice 20: Final challenge

Name: ____________________ Date: __________

Use a table, expression, or equation as helpful.

1. A pattern starts at 9 and adds 6. Write a rule for term n and find term 10.

Work space

2. The rule is $y = 3x + 4$. Find y for $x = 0, 2, 5$.

Work space

3. Sam has 5 more than twice the number of cards Mia has. If Sam has 23, how many does Mia have?

Work space